BiasAperture — Weekly Project Report (WK5)

Period: 2026-08-28 through 2026-09-02
Project: BiasAperture — A Diagnostic and Evaluative Framework for Auditing Demographic Bias in Facial Analysis Systems
Program: Fusemachines AI Fellowship (AIF) 2026 (Kathmandu, Nepal)
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Status: Phase-2 research synthesized · M5 audit/report completion pending

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1. Executive Summary

This reporting period began after the WK4 cutoff on Thursday, August 27. The team completed and merged the new Phase-2 Product Upgrade Sprint research, covering Tracks 21–38, and synchronized the repository's README, project roadmap, research context, and technical summaries with the new findings. The existing diagnostic implementation remains the capstone baseline: FairFace validation inference is complete for 10,954/10,954 images, with the generated predictions CSV ready for the audit step.

Phase 2 remains research-only. It does not change the M1 schema, relax the statistical safeguards, or expand BiasAperture into model retraining or debiasing. The main product direction is now documented, while implementation is gated by the owner decisions recorded in the Phase-2 synthesis.

2. Work Completed

Phase-2 Research

- Merged the Phase-2 synthesis in `research/results/synth_phase2.md`.
- Consolidated findings for Tracks 21 and 24–38 across novelty, UI/UX, modular architecture, deployment/operations, and business/regulatory streams.
- Recorded Track 22 as parked pending Tracks 25 and 36.
- Recorded Track 23 as dropped because cross-modal schema generalization conflicts with the focused vision/face-classifier scope.
- Preserved the additive-only rule: no Phase-2 proposal may modify the locked M1 schema.

Documentation Synchronization

- Updated `README.md` to describe both the original Tracks 01–20 sprint and the Phase-2 Tracks 21–38 sprint.
- Updated the high-level, mid-level, and low-level research summaries to distinguish Phase 1 from Phase 2.
- Added current implementation status, verified data counts, code entry points, and Phase-2 pointers to the paste-first research context.
- Corrected stale explainability descriptions: the current implementation uses demographic-dummy surrogate attribution; spatial/pixel SHAP and ITA skin-tone analysis remain deferred.
- Corrected the WK4 report cutoff to Thursday, August 27 and renamed it to `2026-08-27_WK4_report.md`.
- Added this current-period WK5 report for August 28 through September 2.

Verification and Repository Context

- Confirmed 55 tests are collected under `src/tests/`.
- Confirmed the verified inference output remains `data/processed/fairface_predictions_val.csv`

with 10,954 rows.

- Refreshed Graphify successfully to 1,126 nodes, 1,419 edges, and 144 communities before the latest context-only change.
- A subsequent incremental Graphify update was correctly blocked by the shrink guard after an incomplete semantic pass; the last complete graph was preserved.

3. Current Milestone State

Milestone	Status at period close
M1: Schema lock and baseline	Complete
M2: Data ingestion and test matrix	Complete
M3: Compliance report generation	Complete
M4: Fairness engine and explainability	Complete
M5: Integration and case study	About 90%; validation inference complete, audit report pending
Phase 2: Product Upgrade Sprint	Research synthesis complete; implementation not yet authorized

4. Open Decisions and Next Actions

1. Run the `bias-aperture` audit against `data/processed/fairface_predictions_val.csv` and generate the standalone HTML report.
2. Review DPD, EOD, EOP, DIR, confidence intervals, chi-squared p-values, subgroup sizes, and insufficient-sample handling in the generated report.
3. Finalize empirical tables, confidence intervals, p-values, and disparity visualizations in `report/main.pdf`.
4. Conduct the joint Track 26/27/28 UX review and settle the verdict vocabulary and badge palette.
5. Decide the multi-framework shape for `ReportContext.regulatory_maps` before Track 36 implementation.
6. Verify the source NFR bounds before implementing Track 31 audit profiles.
7. Decide the licensing mechanism before adding any commercial-tier code to the MIT repository.

5. Evidence References

- Phase-2 synthesis: `research/results/synth_phase2.md`
- Phase-2 runner and task map: `research/research tracks/PHASE2_RUNNER_GUIDE.md`, `research/research tracks/PHASE2_TASK_MAP.md`
- Inference completion log: `dev-logs/2026-09-02_session_inference_completion.md`
- Current research context: `research/context feed/CONTEXT.md`
- Previous weekly report: `dev-logs/weekly-reports/2026-08-27_WK4_report.md`